TEXORA · LECTURE SHEET

TQM Tools I: The Seven QC Tools, Six Sigma, Benchmarking and FMEA

The basic quality tools with fully worked textile examples — a garment-defect Pareto analysis and an attribute control chart — plus the new management tools, Six Sigma with a DPMO calculation, benchmarking, and failure mode and effects analysis.

TE-4108 · QUALITY ASSURANCE AND MANAGEMENT IN TEXTILES · TOPIC 3 · LEVEL 4 TERM 1

1. The seven basic QC tools

Ishikawa held that the great majority of workplace quality problems can be attacked with seven basic tools — deliberately simple, usable by front-line teams with paper and pencil:

TOOL	WHAT IT DOES	TEXTILE USE
Check sheet	Structured form for collecting data by category as events occur	Defect tally by type at final inspection
Pareto chart	Bar chart of categories in descending frequency, with cumulative % — separates the vital few from the trivial many	Which 2–3 defect types cause most rejections
Histogram	Distribution of measured values — shape, centre, spread vs tolerance	GSM or yarn count distribution across a lot
Cause-and-effect (fishbone) diagram	Organises brainstormed causes under Man, Machine, Material, Method (and Measurement, Environment)	Why shade variation occurs between batches
Scatter diagram	Plots paired data to reveal correlation	Bath temperature vs ΔE; moisture vs yarn strength
Flow chart	Maps process steps, decisions and rework loops	Exposes hidden rework in the inspection–repair cycle
Control chart	Plots a process statistic over time between statistical control limits — separates common-cause from special-cause variation	Daily defective rate on a p-chart (worked below)

2. Worked example — Pareto analysis of garment defects

A garment factory's final-inspection check sheets for one week record 124 defects in six categories. Sorting descending and accumulating:

DEFECT	COUNT	% OF TOTAL	CUMULATIVE %
Loose thread	46	37.1	37.1
Skipped stitch	31	25.0	62.1
Oil stain	18	14.5	76.6
Seam puckering	12	9.7	86.3
Others (mixed)	9	7.3	93.5
Measurement out of tolerance	8	6.5	100.0
Total	**124**	**100.0**	—

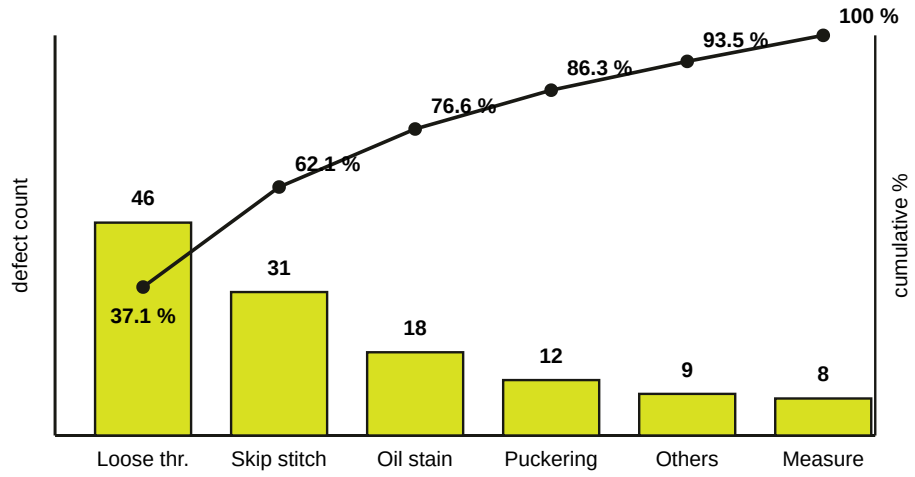


Fig. 1 — Pareto chart of the week's garment defects: bars descending, cumulative line rising to 100 %. The first three categories carry 76.6 % of all defects.

Reading: the vital few are loose thread, skipped stitch and oil stain — 76.6 % of the week's defects from half the categories. Improvement resource goes there first (trimming discipline and thread-path checks; needle and feed maintenance; machine-oil control), not spread thinly over all six. Note the honest bookkeeping: "Others" is plotted in its natural frequency position rather than forced last, since it exceeds the measurement category.

3. Worked example — p-chart for daily defectives

A control chart answers a different question from a Pareto: is the process *stable over time*? For attribute data with samples of n = 200 garments inspected daily, the **p-chart** plots the daily fraction defective p. Ten days of defectives: 12, 9, 14, 8, 11, 10, 15, 9, 12, 24 (total 124 — the same inspection story as Section 2, viewed by day instead of by type).

$$\bar{p} = 124 / (10 \times 200) = 0.062$$

$$UCL = \bar{p} + 3\sqrt{(\bar{p}(1 - \bar{p})/n)} = 0.062 + 3\sqrt{(0.062 \times 0.938 / 200)} = 0.062 + 0.0512 = 0.1132$$

$$LCL = 0.062 - 0.0512 = 0.0108$$

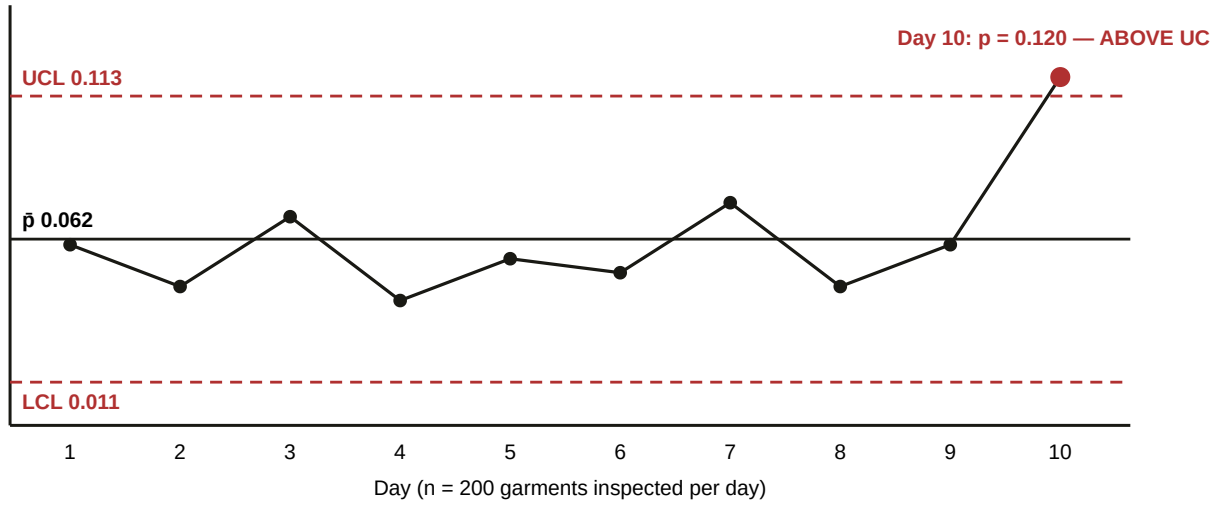


Fig. 2 — p-chart, days 1–10. Days 1–9 vary inside the limits (common-cause variation around p̄ = 6.2 %); day 10 (24 defectives, p = 0.120) exceeds the UCL — a special cause to be found, not a trend to be averaged away.

Interpretation discipline: points inside the limits are the process behaving as it is — improving them requires changing the system (PDCA on the vital-few defects), not exhorting operators about individual days. A point outside signals an assignable cause on that day (a new operator batch, a needle problem, a fabric changeover) — investigate day 10 specifically and promptly, while evidence survives.

4. The new management (planning) tools

Where the seven QC tools handle data, the **seven new management tools** handle language and planning: **affinity diagram** (groups scattered ideas into natural families), **interrelationship diagram** (maps cause–effect links among issues), **tree diagram** (breaks a goal into means, level by level), **matrix diagram** (relates two sets of items — e.g. customer needs vs design features, the heart of QFD in Topic 4), **matrix data analysis** (quantifies matrix relationships), **process decision programme chart (PDPC)** (plans contingencies for what could go wrong), and **arrow/activity network diagram** (schedules interdependent tasks — the family of the project schedules in TE-4107). They are used in planning quality into new products and systems, before data exist for the QC tools.

5. Six Sigma and the DPMO measure

Six Sigma is both a metric and a management system (DMAIC: Define, Measure, Analyse, Improve, Control). The metric counts defects against opportunity: a **unit** is one item; an **opportunity** is each chance for a defect on a unit; DPMO scales the rate to a million opportunities so processes of different complexity can be compared. A process at the "six sigma" benchmark produces 3.4 DPMO (allowing the conventional 1.5σ long-term shift); ordinary industrial processes sit between 3σ (66,807 DPMO) and 4σ (6,210 DPMO).

> **WORKED EXAMPLE — DPMO OF THE GARMENT LINE OF SECTIONS 2–3**
>
> Inspection story: 2,000 garments inspected over the period; 124 defects found; the inspection standard lists 25 defect opportunities per garment (seams, measurements, stains, trims…).
> DPMO = defects ÷ (units × opportunities per unit) × 10^6 = 124 ÷ (2,000 × 25) × 10^6 = 124 ÷ 50,000 × 10^6 = **2,480 DPMO**.
> On the standard sigma table this sits between 4σ (6,210 DPMO) and 4.5σ (1,350 DPMO) — about **4.3σ**: respectable, and far from the 3.4 DPMO six-sigma benchmark.
> Cautions: DPMO rewards defining more "opportunities", so the opportunity list must be fixed by the inspection standard and never padded; and DPMO counts defects, not defectives — one garment may carry several defects, which is exactly why it complements (not replaces) the p-chart's defective rate.

6. Benchmarking

Benchmarking is the systematic comparison of a process or performance metric against a recognised leader — internal (best sister plant), competitive (rival mills), functional (best-in-class anywhere, e.g. a logistics leader for warehouse practice) or generic. The cycle: choose what to benchmark (e.g. claims per million metres), select partners, measure the gap honestly, adapt (not copy) the practices behind the leader's numbers, and re-measure. Benchmarking's quality role is calibration: it stops an organisation grading its own homework — a mill proud of 6 % defectives learns from Fig. 2's chart, but learns its competitive position only when it knows the benchmark plant runs at 2 %.

7. Failure mode and effects analysis (FMEA)

FMEA is preventive: before (or during) a process runs, a team lists each way it could fail (**failure modes**), the **effect** of each failure, its likely **cause**, and scores three 1–10 scales — **Severity (S)** of the effect, **Occurrence (O)** of the cause, **Detection (D)**, where a high D means the failure is hard to detect before escape. The **Risk Priority Number RPN = S × O × D** ranks where to act; actions attack the

highest RPNs (reduce occurrence by error-proofing, improve detection by checks), then RPNs are rescored.

WORKED EXAMPLE — FMEA ROW: WRONG SHADE DYED IN BULK

Failure mode: bulk batch dyed off-shade beyond ΔE tolerance. Effect: buyer claim or redye (Severity 8). Cause: lab-to-bulk recipe correlation drift (Occurrence 5). Current control: first-bulk shade check only at unload (Detection 4 — moderately likely to catch, late).

RPN = 8 × 5 × 4 = **160**.

Actions: re-validate lab-to-bulk correlation monthly (Occurrence → 2); add in-process sample check at 30 minutes into the cycle (Detection → 2).

Revised RPN = 8 × 2 × 2 = **32** — severity cannot be reduced (a wrong-shade batch is always serious), so prevention and detection carry the improvement, as they usually do.

Discipline: a high-severity mode deserves action even at moderate RPN; RPN ranks, it does not excuse.

Glossary

Pareto principle — a small share of categories typically carries most of the effect (the vital few). **Common cause** — variation inherent in the process as designed. **Special (assignable) cause** — variation from an identifiable disturbance, signalled by points outside control limits. **DMAIC** — Six Sigma improvement sequence: Define, Measure, Analyse, Improve, Control. **DPMO** — defects per million opportunities. **Benchmarking** — structured comparison against a leader to find and close performance gaps. **RPN** — risk priority number, S × O × D in FMEA.

Key points

• Seven QC tools: check sheet, Pareto, histogram, fishbone, scatter, flow chart, control chart — simple, front-line, data-first.

• Pareto worked: three defect types = 76.6 % of 124 defects; attack the vital few first.

• p-chart worked: $\bar{p}$ = 0.062, limits 0.1132/0.0108; day 10 (0.120) is a special cause.

• Six Sigma DPMO worked: 2,480 DPMO ≈ 4.3σ for the same inspection story.

• FMEA worked: RPN 160 → 32 by reducing occurrence and improving detection; severity usually fixed.

Practice

1. Defect counts next week: loose thread 38, skipped stitch 27, oil stain 15, puckering 10, others 6, measurement 4 (total 100). Cumulative % after the top three?

2. A p-chart with n = 100/day has $\bar{p} = 0.04$. Compute the control limits. A day with 11 defectives — in or out of control?
3. A weaving department counts 300 defects in 5,000 fabric pieces with 10 opportunities per piece. DPMO?
4. An FMEA row scores S = 9, O = 3, D = 3 (RPN 81) while another scores S = 4, O = 7, D = 6 (RPN 168). Which is acted on first, and what caveat applies?

Answer key

1. Top three total 38 + 27 + 15 = 80 → cumulative 80.0 %. (Full line: 38.0, 65.0, 80.0, 90.0, 96.0, 100.0 %.)
2. $\sigma_p = \sqrt{(0.04 \times 0.96 / 100)} = \sqrt{0.000384} = 0.0196$. UCL = 0.04 + 3 × 0.0196 = 0.0988; LCL = 0.04 − 0.0588 < 0 → LCL = 0 (a fraction cannot be negative; conventionally set to zero). Day with 11 defectives: p = 0.11 > 0.0988 → **out of control** — special cause, investigate.
3. DPMO = 300 ÷ (5,000 × 10) × 10^6 = 300 ÷ 50,000 × 10^6 = **6,000 DPMO** (≈ 4σ band).
4. By RPN ranking, the 168 row is acted on first. Caveat: the 81 row has severity 9 — a near-catastrophic effect when it occurs — so it must not be ignored despite the lower RPN; high-severity modes receive action (typically error-proofing or a fail-safe detection) regardless of rank. RPN prioritises; severity vetoes neglect.
